

1. Table: users

Stores all system users including guests, hosts, and administrators.

Column	Data Type	Key	Nullable	Description
user_id	INT	PK	No	Unique identifier for a user
name	VARCHAR		Yes	Full name of the user
email	VARCHAR		Yes	Email address of the user
role	VARCHAR		Yes	Role of the user (guest, host, admin)
profile_picture	VARCHAR		Yes	URL or reference to the user's profile image
created_at	TIMESTAMP		Yes	Timestamp of when the user account was created

2. Table: booking_statuses

Lookup table containing possible booking statuses.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of booking status
name	VARCHAR		No	Name of the booking status

3. Table: amenities

Lookup table containing amenities available for properties.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of the amenity

name	VARCHAR		No	Name of the amenity
description	VARCHAR		Yes	Description of the amenity

4. Table: payment_methods

Lookup table describing available payment methods.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier for payment method
name	VARCHAR		No	Name of the payment method

5. Table: payment_statuses

Lookup table describing payment transaction statuses.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier for payment status
name	VARCHAR		No	Name of the payment status

6. Table: payment_providers

Stores external payment providers used for processing payments.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier for payment provider
name	VARCHAR		No	Name of the payment provider
contact_email	VARCHAR		Yes	Support contact email for the provider
support_phone	VARCHAR		Yes	Support contact phone number

7. Table: guests

Stores guest-specific user data.

Column	Data Type	Key	Nullable	Description
id	INT	PK, FK	No	Guest identifier referencing users.user_id
phone_number	VARCHAR		No	Phone number of the guest
profile_picture	VARCHAR		Yes	Guest profile image
email	VARCHAR		No	Guest email address

8. Table: hosts

Stores host-specific information.

Column	Data Type	Key	Nullable	Description
id	INT	PK, FK	No	Host identifier referencing users.user_id
phone_number	VARCHAR		No	Phone number of the host
profile_picture	VARCHAR		Yes	Host profile picture
payout_account	VARCHAR		Yes	Payment account used for receiving payouts

9. Table: admins

Stores administrator accounts responsible for managing the platform.

Column	Data Type	Key	Nullable	Description
id	INT	PK, FK	No	Admin identifier referencing users.user_id
phone_number	VARCHAR		Yes	Admin contact phone number
email	VARCHAR		Yes	Admin email address

10. Table: properties

Stores information about rental properties listed by hosts.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of the property
name	VARCHAR		Yes	Name or title of the property listing
description	VARCHAR		Yes	Property description

address	VARCHAR		Yes	Property street address
city	VARCHAR		Yes	City where the property is located
country	VARCHAR		Yes	Country where the property is located
latitude	NUMERIC		Yes	Latitude coordinate of the property
longitude	NUMERIC		Yes	Longitude coordinate of the property
nightly_price	NUMERIC		Yes	Price per night for the property
host_id	INT	FK	No	Host responsible for the property
created_at	TIMESTAMP		Yes	Timestamp when the property was added

11. Table: cancellation_policy

Stores cancellation policies assigned to properties.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of cancellation policy
property_id	INT	FK	No	Property associated with the policy
policy_type	VARCHAR		Yes	Type of cancellation policy
description	VARCHAR		Yes	Description of the cancellation rules

12. Table: property_photos

Stores images associated with properties.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of photo
property_id	INT	FK	No	Property associated with the photo
url	VARCHAR		No	URL or reference to the image
is_primary	BOOLEAN		Yes	Indicates if the photo is the main property image
created_at	TIMESTAMP		Yes	Timestamp when the photo was added

13. Table: property_amenities

Junction table linking properties with available amenities.

Column	Data Type	Key	Nullable	Description
property_id	INT	PK, FK	No	Property identifier
amenity_id	INT	PK, FK	No	Amenity identifier

14. Table: bookings

Stores reservation information created by guests.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of the booking
property_id	INT	FK	No	Property being booked
guest_id	INT	FK	No	Guest making the booking
host_id	INT	FK	No	Host responsible for the property

price	NUMERIC		Yes	Total price of the booking
status_id	INT	FK	No	Status of the booking
start_date	DATE		Yes	Start date of the booking
end_date	DATE		Yes	End date of the booking
created_at	TIMESTAMP		Yes	Timestamp when booking was created

15. Table: disputes

Stores disputes related to bookings.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of dispute
booking_id	INT	FK	No	Related booking
opened_by_user_id	INT	FK	No	User who opened the dispute
admin_id	INT	FK	Yes	Admin assigned to the dispute
reason	VARCHAR		No	Reason for dispute
status	VARCHAR		Yes	Status of the dispute
created_at	TIMESTAMP		Yes	Timestamp when dispute was created
resolved_at	TIMESTAMP		Yes	Timestamp when dispute was resolved

16. Table: messages

Stores messages exchanged between users regarding bookings.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of message
sender_id	INT	FK	No	User sending the message
receiver_id	INT	FK	No	User receiving the message
booking_id	INT	FK	No	Booking associated with the message
parent_message_id	INT	FK	Yes	Parent message if part of a thread
subject	VARCHAR		Yes	Subject of the message
message	VARCHAR		Yes	Message content
created_at	TIMESTAMP		Yes	Timestamp when the message was created

17. Table: payments

Stores payment transactions related to bookings.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of payment
payer_id	INT	FK	No	Guest making the payment
payee_id	INT	FK	No	Host receiving the payment
booking_id	INT	FK	No	Booking related to the payment

amount	NUMERIC		Yes	Payment amount
status_id	INT	FK	No	Payment status
payment_method_id	INT	FK	No	Payment method used
payment_provider_id	INT	FK	No	Payment provider used
created_at	TIMESTAMP		Yes	Timestamp when payment was created

18. Table: refunds

Stores refund transactions related to payments.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of refund
payment_id	INT	FK	No	Payment associated with the refund
amount	NUMERIC		No	Refund amount
created_at	TIMESTAMP		Yes	Timestamp when refund was created
status	VARCHAR		Yes	Refund status

19. Table: booking_cancellations

Stores information about cancelled bookings.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of cancellation
booking_id	INT	FK	No	Booking being cancelled

cancelled_at	TIMESTAMP		No	Timestamp when booking was cancelled
reason	VARCHAR		Yes	Reason for cancellation
refund_amount	NUMERIC		Yes	Refund amount issued for cancellation

20. Table: property_ratings

Stores ratings and reviews for properties submitted by guests.

Column	Data Type	Key	Nullable	Description
id	INT	PK	No	Unique identifier of rating
property_id	INT	FK	No	Property being rated
guest_id	INT	FK	No	Guest submitting the rating
booking_id	INT	FK	No	Booking related to the rating
rating	INT		No	Numeric rating value
description	VARCHAR		Yes	Review text
created_at	TIMESTAMP		Yes	Timestamp when rating was created